

Interactive Digital Art and Design SE608

Project Proposal Year IV



Student Name:	Vit Prechtl	Student Number:	C00298335
Working Title:	Procedural Building Generator		
Description:	A procedural building generator that is game ready and customisable through parameters. The project's primary pillar is Digital Art. The programs that I will be using are: Blender (To model, and unwrap 3d assets, use geometry nodes to use its powerful node-based interface to program with python and a node-based workflow a system that provides an artist parameters to art-direct and design game levels and movie environments, then use blender to set up an environment with the procedural tool, light it, and render it for a presentation), Substance Painter and Affinity (To create textures and texture 3d assets) Core Parameters a user will be able to control (Depending on the process of the project I may add more features, but these are the ones that will be in the tool): - Building width - Building depth - Number of floors - Floor height - Window width - Window height - Window spacing - Window style - Balcony on/off - Balcony size - Roof type - Facade style - Material selection - Detail level - Random seed What will the generator produce: Complete 3D building models in Blender from a set of user-defined parameters. The generated building will include the main building structure, individual floors, facades, windows, doors, optional balconies, and a roof. The system will also allow different materials and architectural styles to be applied to the generated model. When a parameter is changed, the geometry nodes system will automatically regenerate the building, allowing the artist to experiment with different designs		

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	without manually modelling each variation. The generator will also have a random seed control, which will allow the user to create multiple variations of a building while maintaining the same general dimensions and design requirements. This will demonstrate how one procedural system can efficiently produce a range of unique building designs. The final deliverable will consist of a functioning Blender procedural building generator, together with several generated examples which will demonstrate how changes to the parameters affect the final buildings. The technical implementation using Geometry Nodes and Python: Geometry Nodes will be responsible for generating and modifying the building geometry, while Python will be used to create the user interface and manage the generator's parameters inside of Blender viewport side panel. Python will support the geometry nodes system by creating the generator interface which will house the overall product. Python will be essentially the key to turn the geometry nodes template into a blender plugin with its own window.
Reasons for selecting project:	This project has a lot of challenge that will help me to improve myself as a technical artist, by creating a technically complex tool that can be used easily by any artist with the least resistance. It will be also a great tool for an artist to use to focus on their creative and artistic vision, by having a realtime feedback while customising the tool using parameters and being able generate multiple iterations for lots of objects for environments and adjust the generators to their liking, while keeping everything optimised and automatic for an instance based workflow, which is very important in a production pipeline. It can also help a user to save time on hand placing objects in a very mundane way and rather focus on the overall idea of their work.
Proposed research content:	Article about making procedural building generator in Houdini (Chose, because the practices are transferable regardless of the software) https://80.lv/articles/learn-how-to-make-procedural-building-generator-in-houdini In most of the article, the user talks about is creative journey rather than the procedure itself, but I got some great points

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regarding the functionality. (His generator functionalities were aimed at a stylized asset creation. Which in my case I would like to focus more on realism or semi-realism, so I will probably avoid some of the functionalities he had like the lattice and stretch controls)

His generator's functionalities were:

- Adjustable width, depth, and height
- Selectable wall styles for each floor individually
- Option to toggle between just windows or balconies for each floor individually
- Mix of selectable and randomized window styles
- Toggle for a brick overlay
- Slider to control the wear and tear of the roof shingles
- Multiple styles for the main roof
- Bend controls for each roof individually, including range/intensity of bend, and weighting on both sides
- Exposed lattice controls to deform the roof and main building independently.
- Randomized prop scatters for each floor
- Randomized lantern scatter for each floor
- Randomized door seed, with a second door being introduced beyond 5 units wide
- Shrink and stretch controls for the roof, adjustable in both height and width.

Making Procedural Buildings of THE FINALS

<https://www.sidefx.com/community/making-the-procedural-buildings-of-the-finals-using-houdini/>

In this analysis I specifically focus on the overall requirements of a design for a modular system. This article was perfect, since it was directly pulled from a corporate project.

The requirements, considerations or reason for designing a procedural system are:

Creative Freedom – Artists can choose which features a building should have, and combine them in ways that fits their needs.

Scalability – New features can be added over time with minimal risk of breaking existing content.

Maintainability – Since nodes are loosely coupled, bugs or improvements can be often isolated to individual nodes (Making the overall tool more flexible)

Faster iteration – Because of each feature node has well-defined

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dependencies, small changes only recreate the parts of the graph that need it. Adjusting a window won't force a full building to recreate.

Blender Procedural City Generator

<https://www.youtube.com/watch?v=aChmd11adMs&t>

The tool that is advertised in the video is constructed as a blender plugin that lies in the side panel within the Blender viewport. Inside the side panel are the following features that the addon provides.

1. IMPORT CITY

The addon has direct access to the geometry nodes templates, and the geometry nodes are pulled into the scene when a user presses the 'import' button they get a city preset, or when they press the 'append' button inside the addon's side panel the tool will append their tool to their existing object (ex. flat plane). The tool sits as a geometry node modifier on a plane that can be customised in edit mode for the user to design their city based on a ground plan view (meaning the shape of the flat plane). The user can edit the plane using loop cuts or the knife tool, and with that manipulate the look of their city. So the more edges a plane has = more buildings. In short, the plane cuts, and overall mesh manipulation determine the shape of the city.

The generator in this video provides way more than buildings on a plane as it is a city generator, so the user gets buildings, sidewalks, roads, road patches, vegetation, cars, street lights, zebra crossings, and street bins.

2. GEOMETRY SETTINGS

A. The tool provides a proxy preview (low poly preview), that reduces the fidelity of the models to increase viewport performance when making changing adjustments to overall scene.

B. Building Settings

Isolate buildings checkmark – The user can only see the buildings if this feature is checked, which gives the user to just have the buildings or the entire city (Useful when a user already have their own environment, streets and roads)

Height adjustment – There is a slider to setting the min and max height for the buildings. There is also a seed so the user can iterate between the building sizes fast.

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C. Randomize Building Shape

Provides the following sliders: Probability, Min, Max, Seed. All of these together lets the user push the building shape beyond their standard 4 sided shape to a building that has more sides. Min and Max determine the size of the effect, probability will decide how many buildings from the total number of buildings will have this effect, and seed that helps with random iteration.

D. Randomize Z Shape

This feature expands the building on the Z axis (Impacts the height), by layering more elements as skyscrapers in a layers of pyramids. The sliders provide multiple layers, probability slider, subdivision (to make the divisions of the buildings more unique), and a seed.

E. Roof Assets

Sliders for controlling the amount and the distribution of roof assets, like satellittes, electrical units, etc.

F. Facade Settings – Seed to control the asset variation of the facade (The look). Mixture slider that decides how many windows on the buildings are lit up and how many aren't. And finally the density of the fire-escapes.

G. Advanced Settings

All sliders are index inputs that decide the distribution of specific elements that the user wants in the city. For example, how many high-building assets will be in the scene and so on. These index sliders are: high-building assets, fire escapes on facade, amount of different wall assets, and seed.

H. Ground Settings

Adjust the roads and the streets, the distance between the buildings and sidewalks, road markings, manhole covers, zebra markings, road sides, parking cars, and traffic simulation.

3. MATERIAL SETTINGS

A. Road Material

There are parameters that enable the user to change the following:

Slider to change the pattern of the road (Making the texture of the road more irregular), making the road look more new or old. Slider for puddle size, roughness, scale, and distribution.

Slider to adjust the tar (Kinda like road patches), scale, distribution, mask.

Slider to change the color of the roadd markings, and the wear of the road markings.

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	Sliders for the side walk materials to change the sidewalk luminosity, texture, and puddles. Sliders for the brick wall material which are the seed, normal and bump strength (level of texture detail), brightness, contrast, and scale And finally sliders for the variation of a car paint of the cars in the city. 4. LIGHT SETTINGS There are switches for the night lights, car lights.. Then there are settings for the street lights, car lights, and window lights. The settings are the color, size of the light, and brightness.
External links (if applicable):	
Hardware requirements:	Laptop / PC, Mouse, Keyboard
Software requirements:	Blender, Affinity, Adobe Substance 3D Painter
Other requirements:	Textures for the 3D Assets, Concept art to be created

Signed:

Date:

For Office Use Only

Approved/Not Approved:		
Reasons for not approving project:		
Conditions attached to approving project:		
Approved/Not Approved:		
Name of Supervisor:		
Signed: _____		Date: _____

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